

Chapter 5

§5.1

Properties of determinant, denoted by $\det A$ or $|A|$.
 $\downarrow$
 $n \times n$ matrix.

(1) - (16).

In particular.

$$\det(AB) = \det A \cdot \det B \quad \text{but} \quad \det(A+B) \neq \det A + \det B$$

$$\det(A^{-1}) = \frac{1}{\det A}, \quad \det A = \det A^T$$

§5.2.

Big formula. $\det A = \text{sum of all } n! \text{ permutations } P = (\alpha, \beta, \dots, w)$
 $= \sum (\det P) a_{1\alpha} a_{2\beta} \dots a_{nw}$ where $A = (a_{ij})_{n \times n}$.

Cofactor expansion.

$$\det A = a_{i1} C_{i1} + a_{i2} C_{i2} + \dots + a_{in} C_{in}, \quad i = 1, 2, \dots, n.$$

$C_{ij} = (-1)^{i+j} \det M_{ij}$ where M_{ij} is obtained by crossing out row i and column j from A .

or

$$\det A = a_{1j} C_{1j} + a_{2j} C_{2j} + \dots + a_{nj} C_{nj}$$

§5.3.

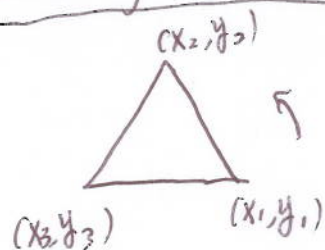
Cramer's Rule If $\det A \neq 0$, then $AX = b$ has a solution

$$x_1 = \frac{\det B_1}{\det A}, \quad \dots, \quad x_n = \frac{\det B_n}{\det A}$$

where B_j has the j -th column of A replaced by b .

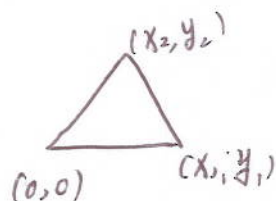
Corollary: $(A^{-1})_{ij} = \frac{C_{ji}}{\det A}$ or $A^{-1} = \frac{C^T}{\det A}$ where $C = (C_{ij})$

Area of a triangle



right-handed complex

$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$



$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$$

Volume of a parallelepiped formed by (a_{11}, a_{12}, a_{13}) , (a_{21}, a_{22}, a_{23}) and (a_{31}, a_{32}, a_{33}) is

$$\text{Volume} = |\det A| = \left| \det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \right|$$

Cross product

$$u \times v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

i) $u \times v = -v \times u$

ii) $u \times u = 0$

iii) $\|u \times v\| = \|u\| \cdot \|v\| \cdot |\sin \theta|$

$$\|u \cdot v\| = \|u\| \cdot \|v\| \cdot |\cos \theta|$$

iv) $\text{Area}(T) = \frac{1}{2} \|u \times v\|$ where T is a triangle formed by u, v .

Triple product

$$(u \times v) \cdot w = \begin{vmatrix} w_1 & w_2 & w_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Chapter 6.

§6.1

A - $n \times n$ matrix

$$AX = \lambda X \quad (X \neq 0)$$

$\downarrow \quad \hookrightarrow$
eigenvalue corresponding eigenvector.

- i) characteristic polynomial $\det(A - \lambda I) = 0 \Leftrightarrow \lambda$ is an eigenvalue.
 $\hookrightarrow$ degree n in λ .
- ii) For each λ , solve $(A - \lambda I)X = 0$ to find corresponding eigenvectors.

Theorem: i) $\det A = \lambda_1 \cdot \lambda_2 \cdots \lambda_n$

ii) $\lambda_1 + \lambda_2 + \cdots + \lambda_n = \text{Trace}(A) = a_{11} + a_{22} + \cdots + a_{nn}$.

Imaginary eigenvalues/eigenvectors — appear in "Conjugate Pairs".

§6.2. Diagonalization of Matrix

Theorem: $S^{-1}AS = \Lambda$ or $A = S\Lambda S^{-1}$ if A has n linearly independent eigenvectors.
where $S = [x_1, x_2, \dots, x_n] \rightarrow$ eigenvector matrix
 $\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} \rightarrow$ eigenvalue matrix

$$AX_i = \lambda X_i, \quad i = 1, 2, \dots, n.$$

Theorem: i) Eigenvectors corresponding to distinct eigenvalues are linearly independent.

ii) A $n \times n$ matrix that has n different eigenvalues must be diagonalizable.

Power of A : $A^k = S \begin{bmatrix} \lambda_1^k & & \\ & \lambda_2^k & \\ & & \ddots \\ & & & \lambda_n^k \end{bmatrix} S^{-1}$ if $A = S\Lambda S^{-1}$

§6.4.

symmetric matrix $A = A^T$

Theorem: i) The eigenvalues of a real symmetric matrix are real.
ii) Eigenvectors of a real symmetric matrix are always perpendicular.

Theorem: Every symmetric matrix A has a complete set of orthogonal eigenvectors:

$$A = S \Lambda S^{-1} \rightarrow A = Q \Lambda Q^T$$

§6.5.

Definition of Positive definiteness of a matrix A .

(symmetric) + $(\lambda_i > 0, i=1, \dots, n) \Rightarrow A$ positive definite.

+ $(\lambda_i \geq 0, i=1, \dots, n) \Rightarrow A$ semidefinite.

Theorem: $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is positive definite $\Leftrightarrow a > 0$ and $ac - b^2 > 0$.

Theorem: The following statements are equivalent for a symmetric matrix A :

- 1) A is positive definite
- 2) All n eigenvalues are positive.
- 3) $X^T A X > 0$ if $X \neq 0$
- 4) All n pivots are positive
- 5) All n upper left determinants are positive.

Cholesky Factorization.

$A = LDL^T$ when A is symmetric.

$$\rightarrow \underline{A = (L\sqrt{D})(L\sqrt{D})^T}$$

Since D is diagonal with positive numbers on the diagonal.